PROJECT: Dual-Homed Network with Dual ISP Connections using Cisco Packet Tracer

QUESTION:

Create a dual-homed network with 2 ISP connections using static and default routes. Configure VLANs internally. Setup DHCP and DNS servers. Add a web server. Practice route metrics to prefer one ISP over another.

OBJECTIVES

- To configure a dual-homed network with two ISP connections using static and default routes in Cisco Packet Tracer.
- To implement VLAN segmentation internally and configure inter-VLAN routing using the Router-on-a-Stick technique.
- To set up DHCP and DNS servers for automatic IP address assignment and domain name resolution across VLANs.
- To deploy a web server and verify end-to-end HTTP connectivity from client devices.
- To practice route metrics to prefer one ISP over another and demonstrate automatic failover when the primary ISP link fails.

THEORY

1. Dual-Homed Networks and ISP Redundancy

A dual-homed network is a network architecture in which a router maintains simultaneous connections to two different Internet Service Providers (ISPs). This provides redundancy and fault tolerance: if one ISP link goes down, traffic is automatically redirected through the second ISP, ensuring continuous connectivity. In this project, the border router R0 connects to both ISP1-R and ISP2-R, simulating a real-world dual-WAN setup.

2. Static Routes and Route Metrics

Static routes are manually configured routing entries that define a fixed path for network traffic. A default static route (0.0.0.0/0) directs all traffic to a specified next-hop gateway when no more specific route exists. Route metrics determine the preference between multiple routes to the same destination. A lower administrative distance or metric means the route is preferred. In this project, the default route via ISP1 is assigned a metric of 1 (primary), while the route via ISP2 is assigned a metric of 10 (backup), forming a floating static route that activates only when the primary route fails.

3. VLANs and Inter-VLAN Routing

A Virtual Local Area Network (VLAN) logically segments a physical network into separate broadcast domains, improving security and traffic management. In this project, three VLANs are configured: VLAN 10 for user PCs, VLAN 20 for servers, and VLAN 30 for management. Since SW1 is a Layer 2 switch, inter-VLAN routing is performed by R0 using the Router-on-a-Stick technique, where a single physical interface is divided into logical sub interfaces, each tagged with 802.1Q encapsulation for a specific VLAN.

4. DHCP and the ip helper-address Command

Dynamic Host Configuration Protocol (DHCP) automates IP address assignment to network clients. When the DHCP server resides on a different VLAN from the clients, DHCP broadcast packets cannot cross VLAN boundaries by default. The ip helper-address command on the router sub interface solves this by forwarding DHCP broadcast requests as unicast packets to the DHCP server's IP address, enabling cross-VLAN DHCP operation.

5. DNS and Web Services

The Domain Name System (DNS) resolves human-readable domain names to IP addresses. In this project, a dedicated DNS server is configured with an A record mapping the domain www.mysite.local to the web server's IP address. The web server runs an HTTP service that responds to browser requests, allowing clients to access the site by name rather than IP address, demonstrating the full application-layer communication stack.

NETWORK TOPOLOGY

The network consists of two ISP routers (ISP1-R and ISP2-R) connected to a border router (R0) via separate WAN links. R0 connects to a Layer 2 switch (SW1) which hosts three VLANs. DHCP, DNS, and Web servers are placed in VLAN 20, while user PCs are in VLAN 10.

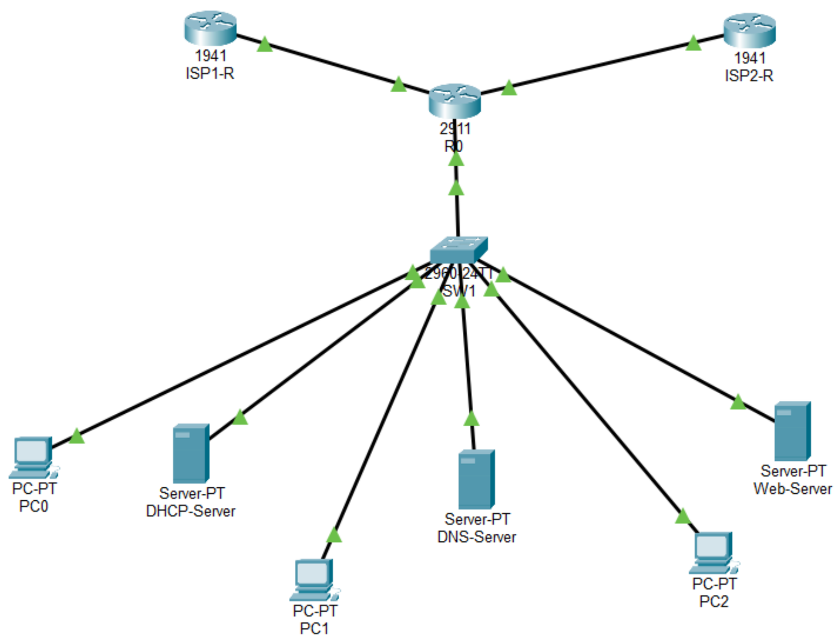


Figure 1: Network Topology

Device	Model	IP Assignment	VLAN / Network	Role
ISP1-R	Cisco 1941	203.0.113.1/30 (Static)	WAN	Simulates ISP 1
ISP2-R	Cisco 1941	198.51.100.1/30 (Static)	WAN	Simulates ISP 2
R0	Cisco 2911	203.0.113.2/30, 198.51.100.2/30	WAN + LAN	Border Router
SW1	Cisco 2960-24TT	N/A	VLAN 10, 20, 30	Layer 2 Access Switch
DHCP-Server	Server-PT	10.1.20.10/24 (Static)	VLAN 20	DHCP Server
DNS-Server	Server-PT	10.1.20.11/24 (Static)	VLAN 20	DNS Server
Web-Server	Server-PT	10.1.20.12/24 (Static)	VLAN 20	HTTP Web Server
PC0	PC-PT	DHCP (10.1.10.x)	VLAN 10	User Client
PC1	PC-PT	DHCP (10.1.10.x)	VLAN 10	User Client
PC2	PC-PT	DHCP (10.1.10.x)	VLAN 10	User Client

VLAN Addressing Plan

VLAN ID	Name	Subnet	Gateway(R0 Sub interface)	Devices
VLAN 10	Users	10.1.10.0/24	10.1.10.1 (Gig0/2.10)	PC0, PC1, PC2
VLAN 20	Servers	10.1.20.0/24	10.1.20.1 (Gig0/2.20)	DHCP, DNS, Web Servers
VLAN 30	Management	10.1.30.0/24	10.1.30.1 (Gig0/2.30)	Management devices

CONFIGURATION

Router R0 Configuration Summary

Parameter	Configured Value
Gig0/0 (WAN - ISP1)	203.0.113.2 / 255.255.255.252
Gig0/1 (WAN - ISP2)	198.51.100.2 / 255.255.255.252
Gig0/2.10 (VLAN 10)	10.1.10.1 / 255.255.255.0, encapsulation dot1Q 10
Gig0/2.20 (VLAN 20)	10.1.20.1 / 255.255.255.0, encapsulation dot1Q 20
Gig0/2.30 (VLAN 30)	10.1.30.1 / 255.255.255.0, encapsulation dot1Q 30
Primary Default Route	ip route 0.0.0.0 0.0.0.0 203.0.113.1 1 (metric 1 - ISP1)
Backup Default Route	ip route 0.0.0.0 0.0.0.0 198.51.100.1 10 (metric 10 - ISP2)
DHCP Helper (VLAN 10)	ip helper-address 10.1.20.10
DHCP Helper (VLAN 30)	ip helper-address 10.1.20.10

SW1 Configuration Summary

Port	Mode	VLAN Assignment
Fa0/1	Trunk	Carries VLAN 10, 20, 30 (uplink to R0)
Fa0/2	Access	VLAN 20 (DHCP-Server)
Fa0/3	Access	VLAN 20 (DNS-Server)
Fa0/4	Access	VLAN 20 (Web-Server)
Fa0/5	Access	VLAN 10 (PC0)
Fa0/6	Access	VLAN 10 (PC1)
Fa0/7	Access	VLAN 10 (PC2)

DHCP Server Configuration Summary

Parameter	Configured Value
Server IP Address	10.1.20.10
Pool Name	VLAN10-Pool
Default Gateway	10.1.10.1
DNS Server	10.1.20.11
Start IP Address	10.1.10.100
Subnet Mask	255.255.255.0
Maximum Users	50

DNS Server Configuration Summary

Parameter	Configured Value
Server IP Address	10.1.20.11
DNS Service	Enabled
A Record Name	www.mysite.local
A Record Address	10.1.20.12

PROCEDURE

Step 1: Build the Network Topology

1. ISP1-R, ISP2-R (Cisco 1941), R0 (Cisco 2911), SW1 (Cisco 2960-24TT), three Server-PT devices, and three PC-PT devices were added to the Packet Tracer workspace.
2. ISP1-R Gig0/0 was connected to R0 Gig0/0 and ISP2-R Gig0/0 was connected to R0 Gig0/1 using copper straight-through cables.
3. R0 Gig0/2 was connected to SW1 Fa0/1. All servers and PCs were connected to SW1 Fa0/2 through Fa0/7 respectively.

Step 2: Configure ISP Routers

1. The CLI of ISP1-R was opened and IP address 203.0.113.1/30 was assigned to GigabitEthernet0/0 with no shutdown.
2. The CLI of ISP2-R was opened and IP address 198.51.100.1/30 was assigned to GigabitEthernet0/0 with no shutdown.

Step 3: Configure Border Router R0

1. IP 203.0.113.2/30 was assigned to Gig0/0 (ISP1 link) and 198.51.100.2/30 was assigned to Gig0/1 (ISP2 link).
2. The IP was removed from Gig0/2 and three subinterfaces (Gig0/2.10, .20, .30) were created with 802.1Q encapsulation and respective VLAN gateway IPs.
3. Two default static routes with different metrics were configured: metric 1 via ISP1 (primary) and metric 10 via ISP2 (backup floating static route).
4. The ip helper-address 10.1.20.10 was added on Gig0/2.10 and Gig0/2.30 to relay DHCP requests to the DHCP server in VLAN 20.

Step 4: Configure VLANs on SW1

1. VLAN 10 (Users), VLAN 20 (Servers), and VLAN 30 (Management) were created using vlan commands in global configuration mode.
2. Fa0/1 was configured as a trunk port to carry all VLANs to R0.
3. Fa0/2, Fa0/3, Fa0/4 were assigned to VLAN 20 (access mode) for servers, and Fa0/5, Fa0/6, Fa0/7 were assigned to VLAN 10 (access mode) for PCs.

Step 5: Configure Servers

1. Static IP 10.1.20.10 was set on DHCP-Server, the DHCP service was enabled, and VLAN10-Pool was created with gateway 10.1.10.1, DNS 10.1.20.11, starting at 10.1.10.100.
2. Static IP 10.1.20.11 was set on DNS-Server, the DNS service was enabled, and an A record was added: www.mysite.local pointing to 10.1.20.12.
3. Static IP 10.1.20.12 was set on Web-Server and the HTTP service was enabled.

Step 6: Configure PCs

1. PC0, PC1, and PC2 were set to DHCP mode under Desktop > IP Configuration.
2. All three PCs received IP addresses in the 10.1.10.100+ range with gateway 10.1.10.1 and DNS 10.1.20.11.

CLI Commands:

Step 1: ISP1-R

```
enable
configure terminal
hostname ISP1-R
interface GigabitEthernet0/0
 ip address 203.0.113.1 255.255.255.252
 no shutdown
exit
exit
write memory
```

Step 2: ISP2-R

```
enable
configure terminal
hostname ISP2-R
interface GigabitEthernet0/0
 ip address 198.51.100.1 255.255.255.252
 no shutdown
exit
exit
write memory
```

Step 3: R0

```
enable
configure terminal
hostname R0
interface GigabitEthernet0/0
 ip address 203.0.113.2 255.255.255.252
 no shutdown
exit
interface GigabitEthernet0/1
 ip address 198.51.100.2 255.255.255.252
 no shutdown
exit
interface GigabitEthernet0/2
 no ip address
 no shutdown
exit
interface GigabitEthernet0/2.10
 encapsulation dot1Q 10
 ip address 10.1.10.1 255.255.255.0
 ip helper-address 10.1.20.10
 no shutdown
exit
interface GigabitEthernet0/2.20
 encapsulation dot1Q 20
 ip address 10.1.20.1 255.255.255.0
 no shutdown
exit
interface GigabitEthernet0/2.30
 encapsulation dot1Q 30
 ip address 10.1.30.1 255.255.255.0
 ip helper-address 10.1.20.10
 no shutdown
exit
ip route 0.0.0.0 0.0.0.0 203.0.113.1 1
ip route 0.0.0.0 0.0.0.0 198.51.100.1 10
exit
write memory
```

Step 4: SW1

```
enable
configure terminal
hostname SW1
vlan 10
 name Users
exit
vlan 20
 name Servers
exit
vlan 30
```

```
 name Management
exit
interface FastEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
exit
interface FastEthernet0/2
 switchport mode access
 switchport access vlan 20
exit
interface FastEthernet0/3
 switchport mode access
 switchport access vlan 20
exit
interface FastEthernet0/4
 switchport mode access
 switchport access vlan 20
exit
interface FastEthernet0/5
 switchport mode access
 switchport access vlan 10
exit
interface FastEthernet0/6
 switchport mode access
 switchport access vlan 10
exit
interface FastEthernet0/7
 switchport mode access
 switchport access vlan 10
exit
exit
write memory
```

Step 5: Verification commands (run on R0)

```
show ip route
show ip interface brief
show running-config
```

Step 6: Verification commands (run on SW1)

```
show vlan brief
show interfaces trunk
show running-config
```

Step 7: Failover test (run on R0)

```
interface GigabitEthernet0/0
 shutdown
show ip route
interface GigabitEthernet0/0
 no shutdown
show ip route
```

OBSERVATIONS

Routing Table Verification

After configuring R0, the show ip route command confirmed the following active routes:

- C 203.0.113.0/30 directly connected via GigabitEthernet0/0 (ISP1 link)
- C 198.51.100.0/30 directly connected via GigabitEthernet0/1 (ISP2 link)
- C 192.168.1.0/24 directly connected via GigabitEthernet0/2 (LAN link to SW1)
- S* 0.0.0.0/0 [1/0] via 203.0.113.1 - ISP1 as the active default route (metric 1)

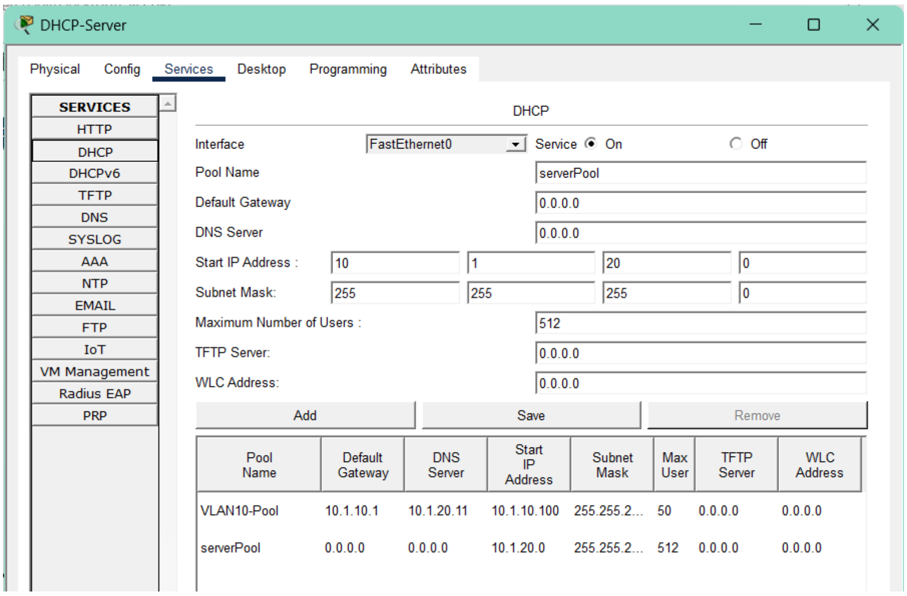
Only one S* entry appeared, confirming that ISP1 was the active preferred path and ISP2 remained as a hidden floating static route with metric 10.

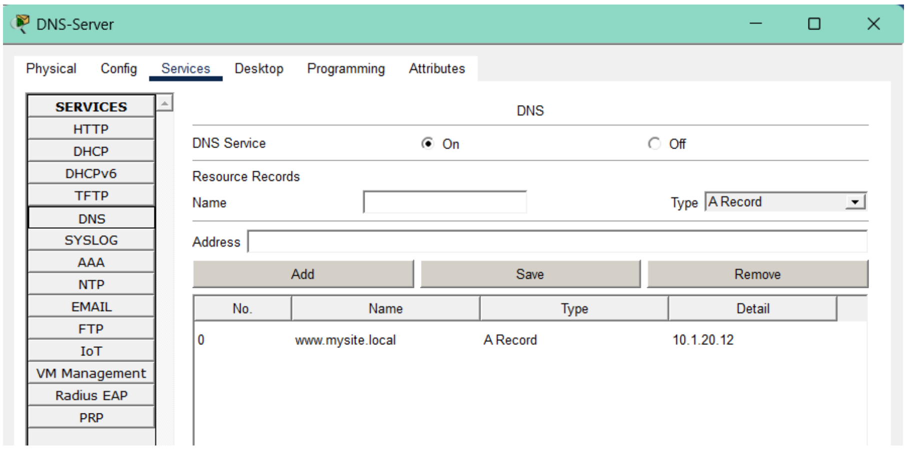
```
R0>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     203.0.113.2     YES manual up                    up
GigabitEthernet0/1     198.51.100.2    YES manual up                    up
GigabitEthernet0/2     unassigned      YES manual up                    up
GigabitEthernet0/2.10  10.1.10.1       YES manual up                    up
GigabitEthernet0/2.20  10.1.20.1       YES manual up                    up
GigabitEthernet0/2.30  10.1.30.1       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R0>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.1.10.0/24 is directly connected, GigabitEthernet0/2.10
L       10.1.10.1/32 is directly connected, GigabitEthernet0/2.10
C       10.1.20.0/24 is directly connected, GigabitEthernet0/2.20
L       10.1.20.1/32 is directly connected, GigabitEthernet0/2.20
C       10.1.30.0/24 is directly connected, GigabitEthernet0/2.30
L       10.1.30.1/32 is directly connected, GigabitEthernet0/2.30
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/30 is directly connected, GigabitEthernet0/1
L       198.51.100.2/32 is directly connected, GigabitEthernet0/1
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/0
L       203.0.113.2/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 203.0.113.1
```

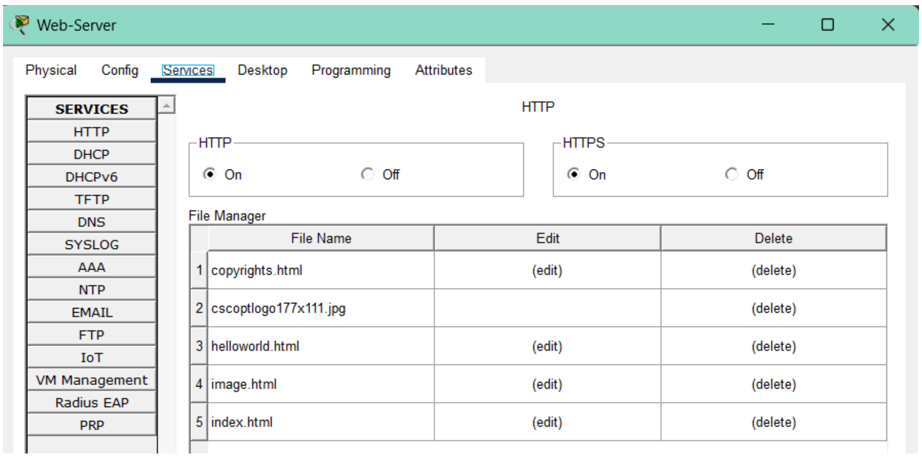
```
SW1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   Users                            active    Fa0/5, Fa0/6, Fa0/7
20   Servers                          active    Fa0/2, Fa0/3, Fa0/4
30   Management                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

DHCP and Cross-VLAN Communication

After configuring the ip helper-address on R0's VLAN 10 sub interface, all three PCs successfully obtained IP addresses via DHCP. Each PC received an address in the 10.1.10.100-102 range, with default gateway 10.1.10.1 and DNS server 10.1.20.11. This confirmed that DHCP relay was functioning correctly across VLAN boundaries.

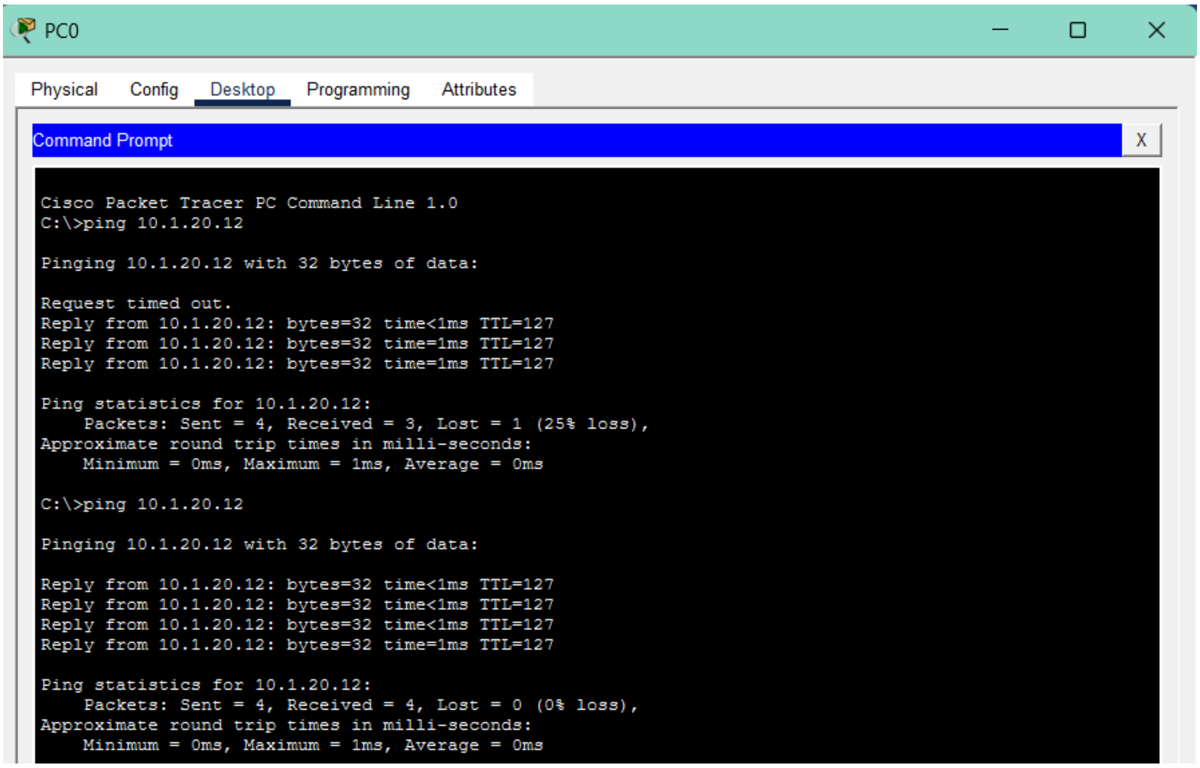


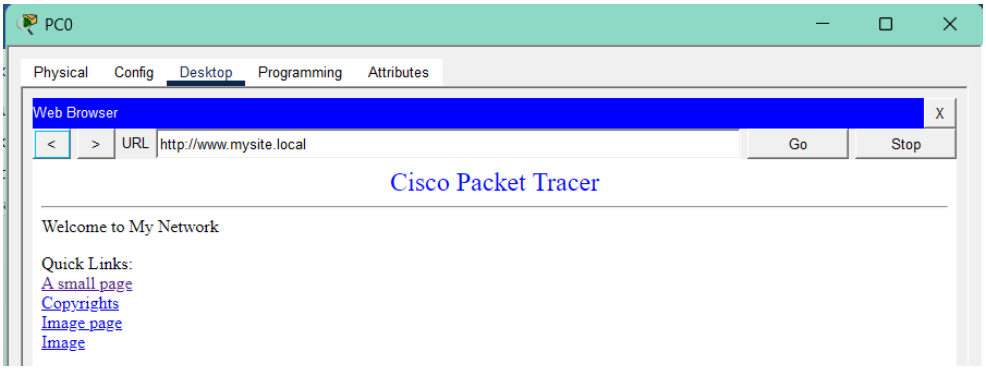




Inter-VLAN Connectivity and Web Access

A ping from PC0 (10.1.10.x in VLAN 10) to the Web-Server (10.1.20.12 in VLAN 20) was successful with 0% packet loss, confirming inter-VLAN routing via Router-on-a-Stick. Opening the web browser on PC0 and navigating to www.mysite.local displayed the custom web page, confirming that DNS resolution and HTTP service were both operational end-to-end.





ISP Failover Test

To test ISP redundancy, GigabitEthernet0/0 on R0 was shut down to simulate an ISP1 failure. Running show ip route again showed the routing table updated automatically: the S* 0.0.0.0/0 entry changed from via 203.0.113.1 (ISP1) to via 198.51.100.1 (ISP2), demonstrating that the floating static route with metric 10 activated as expected. When GigabitEthernet0/0 was restored using no shutdown, traffic reverted to ISP1, confirming automatic failback.

```
R0#enable
R0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#interface GigabitEthernet0/0
R0(config-if)# shutdown

R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
exit
R0(config)#exit
R0#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

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L       10.1.10.1/32 is directly connected, GigabitEthernet0/2.10
C       10.1.20.0/24 is directly connected, GigabitEthernet0/2.20
L       10.1.20.1/32 is directly connected, GigabitEthernet0/2.20
C       10.1.30.0/24 is directly connected, GigabitEthernet0/2.30
L       10.1.30.1/32 is directly connected, GigabitEthernet0/2.30
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/30 is directly connected, GigabitEthernet0/1
L       198.51.100.2/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [10/0] via 198.51.100.1
```

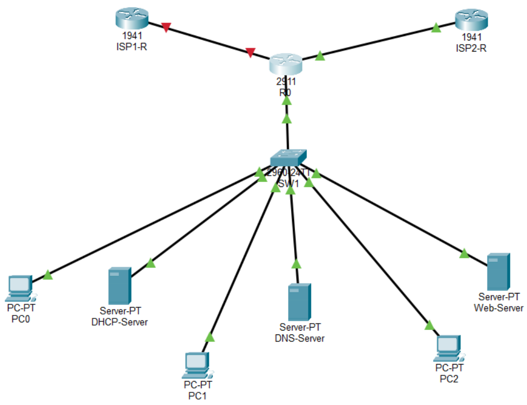


```
C:\>ping 10.1.20.12

Pinging 10.1.20.12 with 32 bytes of data:

Reply from 10.1.20.12: bytes=32 time<1ms TTL=127
Reply from 10.1.20.12: bytes=32 time<1ms TTL=127
Reply from 10.1.20.12: bytes=32 time<1ms TTL=127
Reply from 10.1.20.12: bytes=32 time<1ms TTL=127

Ping statistics for 10.1.20.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

DISCUSSION AND CONCLUSION

This project successfully demonstrated the design and implementation of a dual-homed enterprise network using Cisco Packet Tracer. By connecting a border router to two separate ISPs and assigning different route metrics, a reliable primary/backup WAN architecture was achieved without the need for dynamic routing protocols. The floating static route technique proved effective: ISP2 remained dormant during normal operation and activated seamlessly upon ISP1 failure.

VLAN segmentation divided the internal network into logical groups, improving both security and manageability. The Router-on-a-Stick configuration on R0 enabled inter-VLAN communication over a single physical trunk link, which is a cost-effective approach for small to medium networks using Layer 2 switches. The ip helper-address command was essential for allowing DHCP broadcasts from VLAN 10 to reach the DHCP server in VLAN 20, a common real-world requirement in multi-VLAN environments.

The DNS and web server setup validated application-layer connectivity, confirming that the network stack functioned correctly from physical cabling through to HTTP response. All project objectives were successfully met: dual ISP routing with metric-based preference, VLAN configuration with inter-VLAN routing, DHCP with cross-VLAN relay, DNS resolution, web server access, and ISP failover verification.